On Biadjoint Triangles

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CONTENTS

1.1 Monads and their Algebras

It would be shameful not to begin our exploration without a good background in monads, which are described in detail in this section. Monads, loosely speaking, answer the question of what it means for a category to possess some *algebraic* structure. This quality is made more explicit in the definition of *monadicity*, which cements the idea that a category is *monadic* over another category if it has some algebraic quality in relation to the latter category. This intuition becomes all the more clear once examples of monads and monadicity are studied. Much of this comes directly from Mac Lane, and will begin with a quick remark on composeability of functors and natural transformations.

For a category $\mathbb{X}$ and endofunctor $T : \mathbb{X} \to \mathbb{X}$, there are composite endofunctors $T^2 = T \circ T$ and $T^3 = T^2 \circ T$. If, additionally, there is a natural transformation $\mu : T^2 \to T$, then, for each object x in $\mathbb{X}$, the natural transformations $T\mu$, $\mu T : T^3 \to T^2$ have, respectively, components $(T\mu)_x = T\mu_x$, and $(\mu T)_x = \mu_{Tx}$.

Definition 1.1. [2] A monad $T = (T, \eta, \mu)$ on a category $\mathbb{X}$ is a triple in which $T : \mathbb{X} \to \mathbb{X}$ is an endofunctor on $\mathbb{X}$ and η, μ are two natural transformations

$$1_{\mathbb{X}} \xrightarrow{\eta} T \xleftarrow{\mu} T^2 \tag{1.1}$$

such that the following diagram commutes:

$$\begin{tikzcd} T^3 \arrow[r, "\mu T"] \arrow[d, "T\mu"'] & T^2 \arrow[d, "\mu"'] & T \arrow[l, "\eta T"'] \arrow[d, "T\eta"] \arrow[dl, equal] \\ T^2 \arrow[r, "\mu"'] & T & T^2 \arrow[l, "\mu"] \end{tikzcd} \tag{1.2}$$

Replace in Theorem 1.1 the endofunctor $T : \mathbb{X} \to \mathbb{X}$ with a set M, T^2 and T^3 with the cartesian product $M \times M$ and $M \times M \times M$ respectively, and lastly the natural transformations $\eta : 1_{\mathbb{X}} \to T$ with $\eta : 1 \to M$ and $\mu : T^2 \to T$ with the binary operation $\mu : M \times M \to M$. Then the diagram in Equation (1.2) expresses the usual axioms of a monoid. This is what motivates the naming of the natural transformations η as the identity and μ as the multiplication of a monad (T, η, μ). This is also where we recover the famous "a monad is just a monoid in ..."

Definition 1.2. Let $T = (T, \eta, \mu)$ be a monad on $\mathbb{X}$. A T-algebra is a pair (x, ξ) in which x is an object in $\mathbb{X}$ and $\xi : Tx \to x$ is a morphism in $\mathbb{X}$ making the diagram

$$\begin{tikzcd} T^2x \arrow[r, "\mu_x"] \arrow[d, "T\xi"'] & Tx \arrow[d, "\xi"'] & x \arrow[l, "\eta_x"'] \arrow[dl, equal] \\ Tx \arrow[r, "\xi"'] & x & \end{tikzcd} \tag{1.3}$$

commute. A morphism of T-algebras $h : (x, \xi) \to (x', \xi')$ is a morphism $h : x \to x'$ in $\mathbb{X}$ making the diagram below commute:

$$\begin{tikzcd} Tx \arrow[r, "Th"] \arrow[d, "\xi"'] & Tx' \arrow[d, "\xi'"] \\ x \arrow[r, "h"'] & x' \end{tikzcd} \tag{1.4}$$

Definition 1.3. For a monad $T = (T, \eta, \mu)$ on $\mathbb{X}$, the Eilenberg-Moore category of T-algebras, denoted by T-Alg, is the category of T-algebras with T-algebra morphisms.

Theorem 1.4. *Every adjunction $(F \dashv U, \eta, \epsilon) : \mathbb{X} \to \mathbb{A}$ determines a monad $(UF, \eta, U\epsilon F)$ on $\mathbb{X}$.*

Proof. Firstly, $T = UF$ is an endofucntor on $\mathbb{X}$, η is a natural transformation $\eta : 1_{\mathbb{X}} \to UF = T$ and $\mu = U\epsilon F : UFUF = T^2 \to T = UF$ is a natural transformation. For an object x in $\mathbb{X}$, the commutativity of the diagram below must be checked:

$$\begin{tikzcd} UFUFUFx \arrow[r, "U\epsilon_{FUFx}"] \arrow[d, "UFU\epsilon_{Fx}"'] & UFUFx \arrow[d, "U\epsilon_{Fx}"'] & UFx \arrow[l, "\eta_{UFx}"'] \arrow[dl, equal] \arrow[d, "UF\eta_x"] \\ UFUFx \arrow[r, "U\epsilon_{Fx}"'] & UFx & UFUFx \arrow[l, "U\epsilon_{Fx}"] \end{tikzcd} \tag{1.5}$$

The left square commutes because of the naturality of $\epsilon : FU \to 1_{\mathbb{A}}$ applied to $\epsilon_{Fx} : FUFx \to Fx$, which is preserved under U. The right square commutes because of the triangle identities for the adjunction $(F \dashv U, \eta, \epsilon) : \mathbb{X} \to \mathbb{A}$. The identities are recalled:

$$\begin{tikzcd} F \arrow[r, "F\eta"] \arrow[dr, equal] & FUF \arrow[d, "\epsilon F"] \\ & F \end{tikzcd} \qquad \begin{tikzcd} UFU \arrow[d, "U\epsilon"'] & U \arrow[l, "\eta U"'] \arrow[dl, equal] \\ U & \end{tikzcd} \tag{1.6}$$

This is shown by first considering the bottom triangle in Equation (1.5), which is just the x-component of the left triangle of Equation (1.6) under U, and so it commutes. The top triangle in Equation (1.5) is the Fx-component of the right triangle in Equation (1.5). This shows that $(UF, \eta, U\epsilon F)$ is a monad on $\mathbb{X}$. This monad is called the monad defined by the adjunction $(F \dashv U, \eta, \epsilon) : \mathbb{X} \to \mathbb{A}$. $\square$

Then, naturally, the question arises of whether every monad induces an adjunction. The answer is *yes*, and is discussed in the next theorem.

Theorem 1.5. *Every monad $T = (T, \eta, \mu)$ on $\mathbb{X}$ determines an adjunction $(F^T \dashv U^T, \eta^T, \epsilon^T) : \mathbb{X} \to T$-$\mathsf{Alg}$. The monad defined by this adjunction is the same monad $T = (T, \eta, \epsilon)$.*

Proof. Define

1. the right adjoint $U^T : T\text{-}\mathsf{Alg} \to \mathbb{X}$ to be the forgetful functor, $U^T(x, \xi) := x$;
2. the left adjoint $F^T : \mathbb{X} \to T\text{-}\mathsf{Alg}$ to be the free functor, $F^T x := (Tx, \mu_x)$;
3. the unit $\eta^T : 1_{\mathbb{X}} \to U^T F^T$ to be the same η of the monad;
4. the counit $\epsilon^T : F^T U^T \to 1_{T\text{-}\mathsf{Alg}}$ to be $\epsilon^T_{(Tx,\mu_x)} := \mu_x$.

Now it is checked that this is indeed the desired adjunction. Firstly, note that (Tx, μ_x) is a T-algebra. That is, the diagram from Equation (1.3) commutes:

$$\begin{tikzcd} T^3x \arrow[r, "\mu_{Tx}"] \arrow[d, "T\mu_x"'] & T^2x \arrow[d, "\mu_x"'] & Tx \arrow[l, "\eta_{Tx}"'] \arrow[dl, equal] \\ T^2x \arrow[r, "\mu_x"'] & Tx & \end{tikzcd}$$

But this is just the x-component of Equation (1.2), which commutes because T is a monad.

Now to check naturality of η^T and ϵ^T. The naturality of η^T is just the naturality of η. This is shown by taking x an object in $\mathbb{X}$, then the x-component of η^T is the morphism in $\mathbb{X}$, $\eta^T_x : x \to U^T F^T x = U^T(Tx, \mu_x) = Tx$, which is the same as η_x. Now for ϵ^T: take (x, ξ) an object in T-Alg. Then $F^T U^T(x, \xi) = F^T x = (Tx, \mu_x)$. Then the (x, ξ)-component of ϵ^T is $\epsilon^T_{(x,\xi)} : (Tx, \mu_x) \to (x, \xi)$. Taking a morphism of $h : (x, \xi) \to (x', \xi')$ in T-Alg gives a naturality square:

$$\begin{tikzcd}
(x,\xi) \arrow[d, "h"'] & (Tx, \mu_x) \arrow[r, "\mu_x"] \arrow[d, "F^T U^T h"'] & (x,\xi) \arrow[d, "h"] \\
(x',\xi') & (Tx', \mu_{x'}) \arrow[r, "\mu'_x"'] & (x',\xi')
\end{tikzcd} \tag{1.7}$$

For some morphism $f : x \to x'$ in $\mathbb{X}$, $F^T f : (Tx, \mu_x) \to (Tx', \mu_{x'})$ must be a morphism of T-algebras, which means $F^T f : Tx \to Tx'$ must be defined as $F^T f := Tf$. Then $F^T U^T h = Th$. So $F^T U^T h$ in Equation (1.7) becomes Th, and the square thus commutes by naturality of μ, and becuase h is a morphism of T-algebras. To check that there is an adjunction is to check the triangles in Equation (1.6) commute. For (x, ξ) an object in T-Alg and x and object in $\mathbb{X}$, the left triangle is the (x, ξ)-component of the left triangle of Equation (1.7), and the right is the x-component of the right triangle of Equation (1.7):

$$\begin{tikzcd}
Tx \arrow[r, "T\eta_x"] \arrow[dr, equal] & T^2x \arrow[d, "\mu_x"] \\
& Tx
\end{tikzcd}
\qquad
\begin{tikzcd}
Tx \arrow[d, "\xi"'] & x \arrow[l, "\eta_x"'] \arrow[dl, equal] \\
x &
\end{tikzcd}$$

The left and right traingles commute respectively from the commutativity of the monad diagram Equation (1.2) and the T-algebra diagram Equation (1.3).

Lastly, notice that for an object x in $\mathbb{X}$, $U^T F^T x = Tx$, $\eta^T = \eta$ and $U^T \epsilon^T F^T x = \mu_x$, and so the monad defined by this adjunction is $T = (T, \eta, \mu)$, as desired.

□

So we have that a monad determines an adjunction, which, in return determines the same monad again. The reverse is not always true: an adjunction determines a monad which does not always determine the same adjunction. When an adjunction induces a monad that induces the same adjunction back, we call that adjunction *monadic*. We get a sense of how monadic an adjunction is (or measure its *monadicity* with the following construction.

Theorem 1.6. *Let $(F \dashv U, \eta, \epsilon) : \mathbb{X} \to \mathbb{A}$ be an adjunction and $T = (UF, \eta, U\epsilon F)$ be the monad defined by the adjunction. Then there is a unique functor $K : \mathbb{A} \to T$-Alg with $U^T K = U$ and $KF = F^T$.*

Proof. Define $K : \mathbb{A} \to T$-Alg by $Ka := (Ua, U\epsilon_a)$, and on morphisms $Kf := Uf$. □

There is another arrow that will make the triangle complete: the functor K as defined above has a left adjoint, the functor $L : T$-$\mathsf{Alg} \to \mathbb{A}$.

Theorem 1.7. *There is an adjunction $(L \dashv K, \overline{\eta}, \overline{\epsilon}) : T$-$\mathsf{Alg} \to \mathbb{A}$.*

Proof. We define

1. the left adjoint $L : T\text{-}\mathsf{Alg} \to \mathbb{A}$, where $L(x,\xi)$ is defined as the coequaliser of

$$\begin{tikzcd} FUFx \arrow[r, shift left, "\epsilon_{Fx}"] \arrow[r, shift right, "F\xi"'] & Fx \arrow[r, "\pi_{(x,\xi)}"] & L(x,\xi) \end{tikzcd}$$

2. the right adjoint $K : \mathbb{A} \to T\text{-}\mathsf{Alg}$, defined by $Ka := (Ua, U\epsilon_a)$ on objects and $Kf := Uf$ on morphisms;

3. the unit $\overline{\eta} : 1_{T\text{-}\mathsf{Alg}} \to KL$ defined by $\overline{\eta}_{(x,\xi)} = U\pi_{(x,\xi)}\eta_x$;

4. the counit $\overline{\epsilon} : LK \to 1_{\mathbb{A}}$ is defined for each $\overline{\epsilon}_a$ as the unique morphism making the diagram

$$\begin{tikzcd} FUFUa \arrow[r, shift left, "\epsilon_{FUa}"] \arrow[r, shift right, "FU\epsilon_a"'] & FUa \arrow[r, "\pi_{(Ua,U\epsilon_a)}"] \arrow[dr, "\epsilon_a"'] & (LUa, U\epsilon_a) \arrow[d, "\overline{\epsilon}_a"] \\ & & a \end{tikzcd}$$

   commute.

$\square$

Now our picture is complete, and looks like below, with the F- and U-triangles commuting.

$$\begin{tikzcd} & \mathbb{X} \arrow[dl, shift left, "F"] \arrow[dd, shift right, "F^T"'] \arrow[dd, phantom, "\dashv"] \\ \mathbb{A} \arrow[ur, shift left, "U"] \arrow[ur, phantom, "\top"] \arrow[dr, dashed, shift left, "K"] \arrow[dr, phantom, "\top"] & \\ & T\text{-}\mathsf{Alg} \arrow[uu, shift right, "U^T"'] \arrow[ul, dashed, shift left, "L"] \end{tikzcd} \tag{1.8}$$

Definition 1.8. Let $(F \dashv U, \eta, \epsilon) : \mathbb{X} \longrightarrow \mathbb{A}$ be an adjunction and $T = (T, \eta, \mu)$ be its associated monad. Then

1. the functor $K : \mathbb{A} \longrightarrow T\text{-}\mathsf{Alg}$ as defined in the previous result is called the comparison functor;

2. the functor $U : \mathbb{A} \longrightarrow \mathbb{X}$ is said to be monadic when the comparison functor is an equivalence.

Monadic functors are very useful, because they recall the adjunction from which they came. Moreover, monadicity means that, in some sense, a category is algebraic because it is equivalent to a category of algebras.

As with any free-forgetful adjunction to a category of algebras, it is useful to know what the free algebras of the category are. In the case of the adjunction

Definition 1.9.

Theorem 1.10. *there is an adhybc*

1.2 Beck's Monadicity Theorem

Theorem 1.11 (Beck's Monadicity Theorem)**.** *[2] Let $(F, U, \eta, \epsilon) : \mathbb{X} \longrightarrow \mathbb{A}$ be an adjunction, and $T = (T, \eta, \mu)$ be its associated monad. The following conditions are equivalent:*

1. *the functor $U : \mathbb{A} \longrightarrow \mathbb{X}$ is monadic;*
2. *the functor U preserves coequaliser diagrams of the form:*

$$FUF(X) \underset{F(\xi)}{\overset{\epsilon_{F(X)}}{\rightrightarrows}} F(X) \xrightarrow{\pi_{(X,\xi)}} L(X, \xi)$$

for every T-algebra (X, ξ), and for every object A in $\mathbb{A}$, the diagram

$$FUFU(A) \underset{FU(\epsilon_A)}{\overset{\epsilon_{FU(A)}}{\rightrightarrows}} FU(A) \xrightarrow{\epsilon_A} A$$

is a coequaliser;

3. *U preserves coequaliser diagrams of the form:*

$$FUF(X) \underset{F(\xi)}{\overset{\epsilon_{F(X)}}{\rightrightarrows}} F(X) \xrightarrow{\pi_{(X,\xi)}} L(X, \xi)$$

for every T-algebra (X, ξ), and U reflects isos;

4. *every diagram $A \underset{g}{\overset{f}{\rightrightarrows}} B$ in $\mathbb{A}$ - where $U(A) \underset{U(g)}{\overset{U(f)}{\rightrightarrows}} U(B)$ has an absolute coequaliser - has a coequaliser preserved by U, and U reflects isos.*

Lemma 1.12. *In the situation of Equation* (1.8)*, the functor U^T reflects isomorphisms.*

Proof. Working in the context of Equation (1.8), let (x, ξ) and (x', ξ') be objects in T-Alg such that $(x, \xi) \cong (x', \xi')$. There is an isomorphism $f : x \to x'$ in $\mathbb{X}$, $U^T(x, \xi) = x \cong x' = U^T(x', \xi')$, and so U^T reflects isoss. □

Proof of Theorem 1.11. Consider the situation of Theorem 1.11:

(1) ⇔ (2): Using the labels of Equation (1.8), note that since $U^T K = U$ and U^T reflects isos by Theorem 1.12, then U reflects isos iff so does K. Consider the diagram:

$$\begin{tikzcd}
UFUFx \arrow[r, shift left, "U\epsilon_{Fx}=\mu_x"] \arrow[r, shift right, "UF\xi=T\xi"'] & UFx \arrow[r, "\xi"] \arrow[dr, "U\pi_{(x,\xi)}"'] & x \arrow[d, dashed, "\eta'_x=U\pi_{(x,\xi)}\eta_x"] \\
& & UL(x,\xi)
\end{tikzcd}$$

Let $\mathcal{D}$ be a diagram in $\mathbb{A}$ such that $U(\mathcal{D}) = X \underset{v}{\overset{u}{\rightrightarrows}} X'$ with in $\mathbb{X}$. Suppose $U(\mathcal{D})$ has an absolute coequaliser q.

$$\begin{tikzcd}
T(X) \arrow[r, shift left, "T(u)"] \arrow[r, shift right, "T(v)"'] \arrow[d, "\xi"'] & T(X') \arrow[d, "\xi'"] & \\
X \arrow[r, shift left, "u"] \arrow[r, shift right, "v"'] & X' \arrow[r, "q"'] & Q
\end{tikzcd}$$

Since q is an absolute coequaliser of u and v in $\mathbb{X}$, then $T(q)$ is a coequaliser of $T(u)$ and $T(v)$ in $\mathbb{A}$. Since the u- and v-diagrams commute, then $q\xi'$ also coequalises $T(u)$ and $T(v)$. Then there is a unique morphism $k : T(Q) \longrightarrow Q$ such that $kT(q) = q\xi'$. Since this square commutes, then q is also a morphism in T-Alg. Now we want to show that q is the coequaliser of u and v in T-Alg. Let $q : (X', \xi') \longrightarrow (Q', k')$ be a morphism of T-algebras such that $q'u = q'v$ in T-Alg. Then $q'u = q'v$ in $\mathbb{X}$, and $T(q')T(u) = T(q')T(v)$ in $\mathbb{X}$, which, respectively, induce l and $T(l)$ in the diagram below, by universal property of the coequalisers.

$$\begin{tikzcd}
T(X) \arrow[r, shift left, "T(u)"] \arrow[r, shift right, "T(v)"'] \arrow[dd, "\xi"'] & T(X') \arrow[r, "T(q)"] \arrow[drr, "T(q')"'] \arrow[dd, "\xi'"] & T(Q) \arrow[dr, dashed, "T(l)"] \arrow[dd, dashed, "k"] & \\
 & & & T(Q') \arrow[dd, "k'"] \\
X \arrow[r, shift left, "u"] \arrow[r, shift right, "v"'] & X' \arrow[r, "q"] \arrow[drr, "q'"'] & Q \arrow[dr, dashed, "l"] & \\
 & & & Q'
\end{tikzcd}$$

Now we just need to show that l is also a morphism of T-algebras, which is to show that $k'T(l) = lk$. Consider a segment of the diagram:

$$\begin{tikzcd}
T(X') \arrow[r, "T(q)"] \arrow[d, "\xi'"'] & T(Q) \arrow[r, dashed, "T(l)"] \arrow[d, "k"] & T(Q') \arrow[d, "k'"] \\
X' \arrow[r, "q"'] & Q \arrow[r, dashed, "l"'] & Q'
\end{tikzcd}$$

The outer square commutes because $k'T(l)T(q) = k'T(q') = q'\xi' = lq\xi'$. The left square also commutes by assumption, and T(q) is an epi since it is a coequaliser. Then by a pasting lemma, the right square commutes, too. Then l is indeed a T-algebra morphism, which shows that q is the coequaliser of u, v in T-Alg. From (1), T-$\mathsf{Alg} \sim \mathbb{A}$, which means there is also a coequaliser of the diagram $\mathcal{D}$ in $\mathbb{A}$, which is trivially preserved by U. □

Now we want a simpler way to check monadicity, which will become useful in the next example.

Theorem 1.13. *Crude Monadicity Theorem Let $U : \mathbb{A} \longrightarrow \mathbb{X}$ be a functor. U is monadic if*

1. *U has a left adjoint;*
2. *$\mathbb{A}$ has, and U preserves, reflexive coequalisers;*
3. *U reflects isomorphisms.*

Proof. Let us only prove the second point. □

1.3 The Adjoint Triangle Theorem

Definition 1.14. [1] Given a functor $\mathbb{A} \xrightarrow{J} \mathbb{B}$, an adjoint triangle for J consists of a category $\mathbb{C}$, and adjunctions

$$(E \dashv R, \rho, \mu) : \mathbb{A} \to \mathbb{C}, \quad (L \dashv U, \eta, \varepsilon) : \mathbb{B} \to \mathbb{C} \tag{1.9}$$

$$\rho : \mathsf{id}_{\mathbb{A}} \to RE, \quad \eta : \mathsf{id}_{\mathbb{B}} \to UL \tag{1.10}$$

$$\mu : ER \to \mathsf{id}_{\mathbb{C}}, \quad \epsilon : LU \to \mathsf{id}_{\mathbb{C}} \tag{1.11}$$

with $LJ = E$, displayed in the diagram:

$$\begin{tikzcd}
\mathbb{A} \arrow[rr, "J"] \arrow[dr, shift right, "E"'] & & \mathbb{B} \arrow[dl, shift left, "L"] \\
& \mathbb{C} \arrow[ul, shift right, "R"'] \arrow[ur, shift left, "U"] &
\end{tikzcd}$$

. The equality $LJ = E$ is sometimes written as the natural transformation $LJ \xrightarrow{1} E$ and *vice verse.*

Lemma 1.15. *[1] In the notation above, given an adjoint triangle for a functor J, there is a natural transformation $JR \xrightarrow{\Sigma} U$ which makes the following diagram commute:*

$$\begin{tikzcd}
JRLJ \arrow[r, "\Sigma LJ"] \arrow[d, equals] & ULJ \\
JRE & J \arrow[l, "J\rho"] \arrow[u, "\eta J"']
\end{tikzcd}$$

Proof. Define Σ as the composite:

$$\begin{tikzcd}
ULJR \arrow[r, equals] & UER \arrow[d, "U\mu"] \\
JR \arrow[u, "\eta JR"] \arrow[r, "\Sigma"'] & U
\end{tikzcd}$$

Then we fill in the data of the diagram:

$$\begin{tikzcd}
JRLJ \arrow[rrrrrr, "\Sigma LJ"] \arrow[dr, "\eta JRLJ"'] & & & & & & ULJ \\
& ULJRLJ \arrow[rr, "U1RLJ"] & & UERLJ \arrow[urrr, "U\mu LJ"'] & & & \\
& & & & & UE \arrow[uur, "U1"'] \arrow[d, equals] & \\
& & UERE \arrow[uur, "UER1"] \arrow[urrr, "U\mu E"] & & & UE \arrow[lll, "UE\rho"] & \\
& ULJRE \arrow[uuu, "ULJR1"] \arrow[ur, "U1RE"] & & & & ULJ \arrow[llll, "ULJ\rho"'] \arrow[u, "U1"] \arrow[uuuur, equals] & \\
JRE \arrow[uuuuu, "JR1"] \arrow[ur, "\eta JRE"] & & & & & & J \arrow[llllll, "J\rho"'] \arrow[ul, "\eta J"] \arrow[uuuuu, "\eta J"']
\end{tikzcd}$$

$\square$

Theorem 1.16 (Adjoint Triangle Theorem)**.** *[1] Consider an adjoint triangle for a functor $J : \mathbb{A} \to \mathbb{B}$ using the notation of Theorem 1.14. Assume that η is the equaliser of the diagram*

$$\mathit{id}_{\mathbb{B}} \xrightarrow{\eta} UL \underset{\eta UL}{\overset{UL\eta}{\rightrightarrows}} ULUL$$

. If the equaliser of the diagram

$$\begin{tikzcd} RL \arrow[rr, "RL\eta"] \arrow[dr, "\rho RL"'] & & RLUL \\ & RERL = RLJRL \arrow[ur, "RL\Sigma L"'] & \end{tikzcd}$$

exists in the category $\mathbb{A}^{\mathbb{B}}$, then J has a right adjoint G, and G is computed as this equaliser.

Now we put these things together. For an object Y in $\mathbb{B}$:

$$\rho$$

Because the Yoneda embedding preserves limits, the *Adjoint Triangle Theorem* may equivalently be formulated in terms of Hom-sets.

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